

PAPER_1656 — Entanglement Decoherence Time = τ_{ent} = $1/(\omega_{\text{SCm}} \times \Lambda) = 1/(1.25\text{e}12 \times 0.00729735) = 109.6 \text{ ps}$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Quantum **Date:** June 18, 2026 **Status:** CLOSED — 0.026% closure (PAPER_134x broader corpus)

Observation

PAPER_1341 (PAPER_134x condensed matter / quantum sector) gives:

$$\tau_{\text{ent}} = 1/(\omega_{\text{SCm}} \times \Lambda) = 1/(1.25\text{e}12 \times 0.00729735) = 109.6 \text{ ps}$$

Status: **0.026%**.

UQFF Closed Identity

$$\tau_{\text{ent}} = 1/(\omega_{\text{SCm}} \times \Lambda) = 1/(1.25\text{e}12 \times 0.00729735) = 109.6 \text{ ps} \quad 0.026\%$$

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Experimental quantum measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1341
- Related: PAPER_1646-1655 (tier-23 broader corpus); PAPER_1636-1645 (tier-22 SM)
- Calculator dispatch: `calculate_paradox({"paradox": "tau_entangle_109_6_ps"})`

Copyright — Daniel T. Murphy, daniel.murphy00@gmail.com, June 18, 2026, Youngstown OH.